

Fady Nabil

AI Engineer

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Github: github.com/FadyNF · Portfolio: portfolio-seven-drab-oanh6165ps.vercel.app

EDUCATION

Misr International University

Bachelor in Computer Science (AI Minor) — GPA: 3.16 / 4.0

Cairo, Egypt

2022 – 2026 (Expected)

EXPERIENCE

ElSewedy Electric Headquarters

Artificial Intelligence Intern

Cairo, Egypt

Jul. 2025 – Aug. 2025

- Built a production HR chatbot using a hand-rolled RAG pipeline (PostgreSQL + pgvector, OpenAI API, Streamlit), achieving 100% policy query coverage across the company's full HR documentation.
- Engineered a semantic PDF chunking system using the unstructured library and OCR, cutting retrieval latency by structuring document splits around meaning rather than fixed token windows.

Al Ahli Bank of Kuwait — Egypt Headquarters

Applications Developer Intern

Cairo, Egypt

Aug. 2024 – Sep. 2024

- Built a loan and lease payment scheduling system in C# and SQL Server, automating end-to-end payment tracking for a banking operation processing hundreds of transactions monthly.
- Led requirements gathering through stakeholder interviews, translated business rules into system logic, and delivered the full implementation solo within a 1-month internship window.
- Automated payment tracking and scheduling workflows, reducing manual processing errors by 80% and eliminating an estimated 15+ hours of weekly manual work.

Misr International University

Teaching Assistant

Cairo, Egypt

2023 – 2025

- Co-assisted in 4 core CS courses: Intro to CS, Computer Programming, Data Structures & Algorithms, and OOP — supporting students in labs, grading, and concept reinforcement.

PROJECTS

ReVerie — VR Therapeutic Platform for Alzheimer's Patients | Flutter, GCP, Supabase, FastAPI, Unreal Engine

Sep. 2025 – Current

- Led system architecture and integration of a VR-based Alzheimer's therapy platform combining cognitive stimulation therapy with AI-generated 3D environments, built across a 5-person team.
- Integrated the full AI pipeline: a 7-stage image-to-3D reconstruction system (depth estimation, segmentation, object + human reconstruction, etc.) averaging ~10–15 min per scene on GCP L4 GPU.
- Migrated all AI services and backend infrastructure to GCP, reducing local setup overhead by ~70% and enabling parallel AI inference across the team.
- Built the Flutter mobile app end-to-end (Riverpod, Supabase, Agora video, biometric auth) and integrated it with the AI pipeline, backend APIs, and Unreal Engine VR system.

Extractive Text Summarization App | NLP, HuggingFace, BERT, BART, T5, BiLSTM, XGBoost

May 2025 – Jun. 2025

- Built and benchmarked 9 summarization models (KNN, Logistic Regression, XGBoost, BERTSum, BiLSTM, LSTM, MiniLM, BART, T5) across BBC and IMDB datasets.
- Achieved 100% accuracy and ROUGE-L 0.886 on BBC (KNN), and 92% accuracy with ROUGE-L 0.709 on IMDB (Logistic Regression) — outperforming several transformer baselines on extractive tasks.

Football Analysis System | Deep Learning, YOLOv8, ByteTrack, KMeans, Python

Jan. 2025

- Built a real-time football analysis system using YOLOv8 + ByteTrack for multi-player detection and tracking across full match videos.
- Applied KMeans clustering on jersey color features to automatically assign players to teams — no manual labeling required.

More projects available on my [portfolio](#).

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, PHP, JavaScript, SQL, Dart (Flutter)

AI / ML: PyTorch, TensorFlow, HuggingFace Transformers, NLP Pipelines, Computer Vision, 3D Reconstruction, YOLO, OpenCV, RAG, LLMs, Prompt Engineering

Backend & Databases: FastAPI, Node.js, Express.js, PostgreSQL, MongoDB, Supabase, pgvector, REST API Design

Cloud & DevOps: GCP, Azure, Linux, Git / GitHub

Web: HTML, CSS, JavaScript, PHP

RESEARCH

AI-Driven Digital Therapeutic Interventions for Early-Stage Alzheimer's Disease: A Systematic Literature Review (Under Review)

- Submitted to *Complexity* (Wiley), 2025. Reviewed AI-driven memory reconstruction, VR reminiscence therapy, computer vision, digital twins, and cognitive stimulation for early-stage Alzheimer's.

ReVerie: Personalized AI-Based 2D-to-3D Memory Reconstruction for Immersive Reminiscence Therapy (Under Review)

- Submitted to the MIU Computer Conference (MIUCC'26), 2026. Proposed a personalized 2D-to-3D reconstruction pipeline transforming patient photographs into immersive VR scenes.

Comparative Analysis of Machine Learning and Deep Learning Techniques for Extractive Text Summarization

- Contributed paper at the 5th International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC'25), IEEE Egypt Section, Sep. 2025. Benchmarked 9 ML/DL models across BBC and IMDB datasets.

CERTIFICATIONS

- IBM SkillsBuild — [AI Fundamentals with Capstone](#) (Feb 2025)
- IBM SkillsBuild — [Generative AI for Software Development](#) (Feb 2025)
- IBM SkillsBuild — [Generative AI for Data Insights](#) (Feb 2025)